Code No: 126AQ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
INFORMATION SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a) What are the possible attacks on computers? [2]
b) Differentiate between encryption and decryption. [3]
c) What is differential cryptanalysis? [2]
d) Explain Blowfish algorithm in brief. [3]
e) Define Message Authentication code. [2]
f) What is the function of TGS Server in Kerberos? [3]
g) What are different approaches to Public-key Management? [2]
h) List the different encryption and authentication algorithms used for AH and ESP protocols. [3]
i) What is Secure Socket Layer? [2]
j) What are the limitations of firewalls? Explain. [3]

PART - B

(50 Marks)

2.a) Explain various security services and mechanisms.
b) Discuss about symmetric and asymmetric key cryptography. [5+5]

OR

3.a) Explain the basic model for network security.
b) List and explain substitution and transposition techniques in brief. [5+5]

4.a) Differentiate between the symmetric block ciphers and symmetric stream ciphers.
b) Explain the process encryption and decryption in Elliptic curve cryptography. [5+5]

OR

5.a) What are the principles of public key cryptosystems? Discuss.
b) Explain Diffie-Hellman Key Exchange in Elliptic curve cryptography. [5+5]

6.a) Explain about X.509 authentication service.
b) What are the approaches used for Digital Signatures based on Public Key Encryption? Discuss. [5+5]

OR

7.a) Discuss in detail about whirlpool algorithm.
b) With a neat sketch, explain the concept of Secured Hash Algorithm (SHA). [5+5]

8. Draw the IP security authentication header and explain the functions of each field. [10]

OR

9.a) What are the five principal services provided by PGP? Explain.

b) How the messages are generated and transmitted in Pretty Good Privacy (PGP) protocol? Explain. [5+5]

10.a) What are the steps involved in performing Secure Inter-branch Payment Transactions? Explain.

b) Discuss in detail about various password management techniques. [5+5]

OR

11.a) What are the various virus counter measures? Discuss.

b) List the characteristics of a good firewall implementation. How is circuit gateway different from application gateway? [5+5]

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